

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

**Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation |
Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry | Unit 10 — Practical Geometry**

Total Marks: 65

Time Allowed: 2 Hours 30 Minutes

Version: 1

General Instructions:

1. This paper has three sections: Section A (MCQs), Section B (Short Questions), and Section C (Long Questions).
2. Attempt Section A on the question paper by filling the bubble next to the correct option.
3. In Section B, attempt any 11 questions. For each, choose either the main question **OR** its alternative.
4. In Section C, attempt all 4 long questions. For each, choose either the main question **OR** its alternative.
5. Use black or blue ink only. Pencil is allowed only for diagrams and geometric constructions. Show all working.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which of the following numbers is irrational?	$\sqrt{7}$	$\sqrt{16}$	$22/7$	$0.\bar{7}$	ⒶⒷⒸⒹ
2	The simplest radical form of $\sqrt{50}$ is:	$25\sqrt{2}$	$5\sqrt{2}$	$2\sqrt{5}$	$10\sqrt{5}$	ⒶⒷⒸⒹ
3	The value of $\log_{10} 1000$ is:	2	10	3	4	ⒶⒷⒸⒹ
4	By the laws of logarithms, $\log(mn)$ is equal to:	$\log m - \log n$	$\log m + \log n$	$(\log m)(\log n)$	$n \log m$	ⒶⒷⒸⒹ
5	The factors of $x^2 - 7x + 12$ are:	$(x - 3)(x - 4)$	$(x + 3)(x + 4)$	$(x - 6)(x - 2)$	$(x - 12)(x + 1)$	ⒶⒷⒸⒹ
6	The factorization of $a^3 + 8$ is:	$(a - 2)(a^2 + 2a + 4)$	$(a + 2)(a^2 + 2a + 4)$	$(a - 2)(a^2 - 2a + 4)$	$(a + 2)(a^2 - 2a + 4)$	ⒶⒷⒸⒹ
7	The solution of the equation $2x + 3 = 11$ is:	$x = 5$	$x = 7$	$x = 4$	$x = 8$	ⒶⒷⒸⒹ
8	The solution set of $ x = 5$ is:	{5}	{-5}	{ } (empty)	{-5, 5}	ⒶⒷⒸⒹ
9	The distance between the points (0, 0) and (3, 4) is:	7	5	25	1	ⒶⒷⒸⒹ
10	The midpoint of the segment joining (2, 4) and (6, 10) is:	(3, 7)	(4, 6)	(8, 14)	(4, 7)	ⒶⒷⒸⒹ
11	The line segment joining the centre of a circle to any point on it is called the:	Radius	Chord	Diameter	Tangent	ⒶⒷⒸⒹ
12	The minimum number of independent measurements needed to construct a unique triangle is:	1	2	3	4	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. For each pair, attempt either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	(a) Simplify $\sqrt{72}$ to its simplest radical form. (b) Rationalize the denominator: $3 / \sqrt{2}$.	3	OR	State whether each number is rational or irrational, with a reason: (a) $\sqrt{36}$ (b) $\sqrt{10}$ (c) $0.\bar{7}$	3
2(ii)	Simplify using the laws of exponents: (a) $(3^2 \times 3^4) \div 3^3$ (b) $(a^{-2})^{-3}$ (give a positive index).	3	OR	Express in standard scientific notation: (a) 0.00046 (b) 92,000,000.	3
2(iii)	Using the laws of logarithms, evaluate: (a) $\log_2 32$ (b) $\log_{10} 1000 - \log_{10} 10$.	3	OR	Write in expanded form using the laws of logarithms: $\log(a^2b/c)$.	3
2(iv)	Given $\log 2 = 0.3010$ and $\log 3 =$	3	OR	Find the value of x: (a) $\log_3 x = 4$ (b)	3

	0.4771, find: (a) $\log 6$ (b) $\log 12$.			$\log_x 81 = 4$.	
2(v)	Factorize: (a) $x^2 - 9x + 20$ (b) $2x^2 + 7x + 3$.	3	OR	Factorize: (a) $8a^3 + 27$ (b) $64x^3 - 1$.	3
2(vi)	Find the HCF of $12x^2y^3$ and $18x^3y^2$ by factorization.	3	OR	Find the LCM of $x^2 - 1$ and $x^2 + 2x + 1$.	3
2(vii)	Solve for x: $(2x)/3 + 1 = x/2 + 2$.	3	OR	Solve the absolute value equation: $ 2x + 5 = 11$.	3
2(viii)	Solve the inequality and write its solution set: $8x + 9 < 6x - 7$.	3	OR	Solve the inequality and write its solution set: $2(x - 3) \geq 4x + 2$.	3
2(ix)	Find the distance between the points A(-2, 1) and B(1, 5).	3	OR	Find the distance between the points P(3, -2) and Q(-1, 1).	3
2(x)	Find the midpoint of the segment joining (2, 5) and (6, 9), and state the quadrant in which it lies.	3	OR	If M(1, 4) is the midpoint of the segment joining A(6, b) and B(a, 2), find a and b.	3
2(xi)	Define the following terms used in practical geometry: (a) chord (b) tangent to a circle (c) arc.	3	OR	Write the steps to bisect a given line segment AB using only a compass and a straightedge.	3

SECTION C — Long Questions ($4 \times 5 = 20$ Marks) — Attempt all 4

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	(a) Factorize: $(x + 1)(x + 2)(x + 3)(x + 4) - 24$. [3] (b) Find the square root of $4x^2 + 12xy + 9y^2$ by factorization. [2]	5	OR	(a) Find the square root of $x^4 + 8x^3 + 20x^2 + 16x + 4$ by the division method. [3] (b) Find the HCF of $x^2 - 5x + 6$ and $x^2 - 9$ by factorization. [2]	5
3(ii)	(a) State any three laws of logarithms. [2] (b) Given $\log 2 = 0.3010$, $\log 3 = 0.4771$, $\log 7 = 0.8451$, evaluate: (i) $\log 42$ (ii) $\log (9/2)$. [3]	5	OR	(a) Define common logarithm and natural logarithm; define the characteristic and mantissa of a logarithm. [2] (b) Find the value of x: (i) $\log_{10} x = 2$ (ii) $\log_2(x - 1) = 3$ (iii) $\log_5 125 = x$. [3]	5
3(iii)	(a) Solve and verify your answer: $(x - 2)/3 + (x + 1)/2 = 4$. [3] (b) Solve and represent the solution set on a number line: $3x - 5 \leq 7$, $x \in \mathbb{R}$. [2]	5	OR	(a) Solve the absolute value equation: $ 3x - 2 = x + 4 $. [3] (b) Solve the compound inequality and write the solution set: $-3 < 2x + 1 \leq 7$, $x \in \mathbb{R}$. [2]	5
3(iv)	(a) Show that the points A(-4, 2), B(1, 4), C(3, -1), D(-2, -3) are the vertices of a square. [4] (b) State the distance formula. [1]	5	OR	(a) Using a ruler and compass, construct a triangle ABC in which $AB = 5$ cm, $BC = 6$ cm and the included angle $\angle B = 60^\circ$. Write the steps of construction. [3] (b) Describe how to draw the perpendicular bisector of a line segment of length 6 cm. [2]	5

Invigilator's Signature: _____ Candidate's Name: _____ Roll No: _____

Units: Unit 1 (Real Numbers) | Unit 2 (Logarithms) | Unit 4 (Factorization & Algebraic Manipulation) | Unit 5 (Linear Equations & Inequalities) | Unit 7 (Coordinate Geometry) | Unit 10 (Practical Geometry)

— End of Question Paper —